

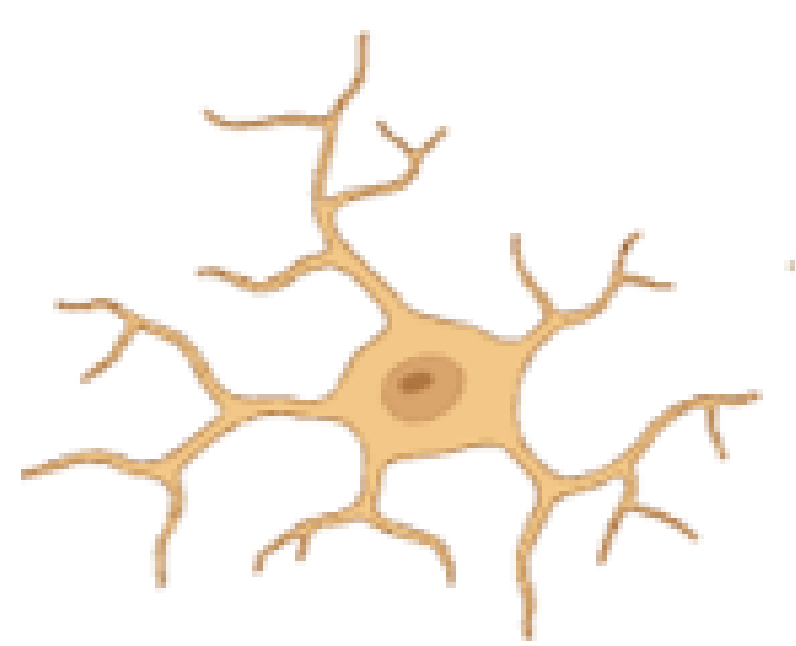
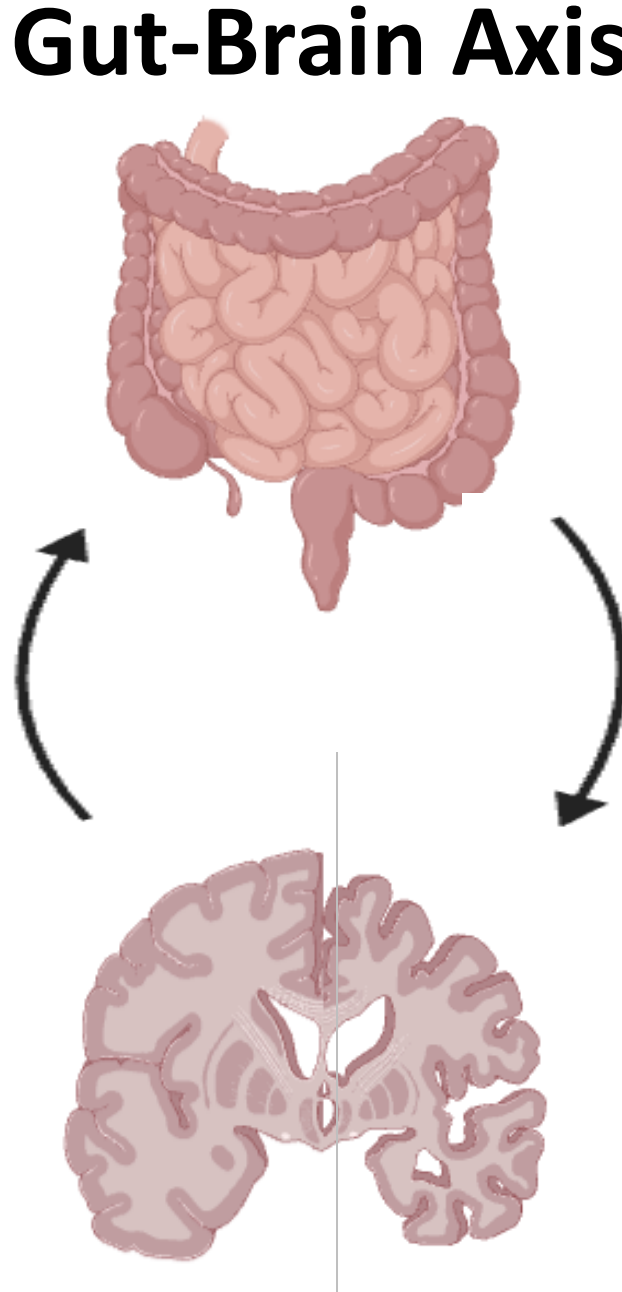
Background

- Inflammatory Bowel Disease (IBD)**
 - Altered gut microbiome
 - Systemic and intestinal inflammation
 - Varied microbiota metabolite production
- Alzheimer’s Disease (AD)**
 - Common neurodegenerative disease and type of dementia
 - Characterized by **amyloid-beta plaques** and **neurofibrillary tangles**

People with **IBD** are **six times** more likely to develop **AD pathology**¹

- Microglia**
 - Resident immune cell** of the brain
 - Responds to gut inflammation
 - Regulates plaque content*

*Hypothesis: IBD microbiome alters standard **microglial activity** through inflammation and altered metabolite production, causing IBD mice models to have **higher susceptibility to AD pathology** development.*

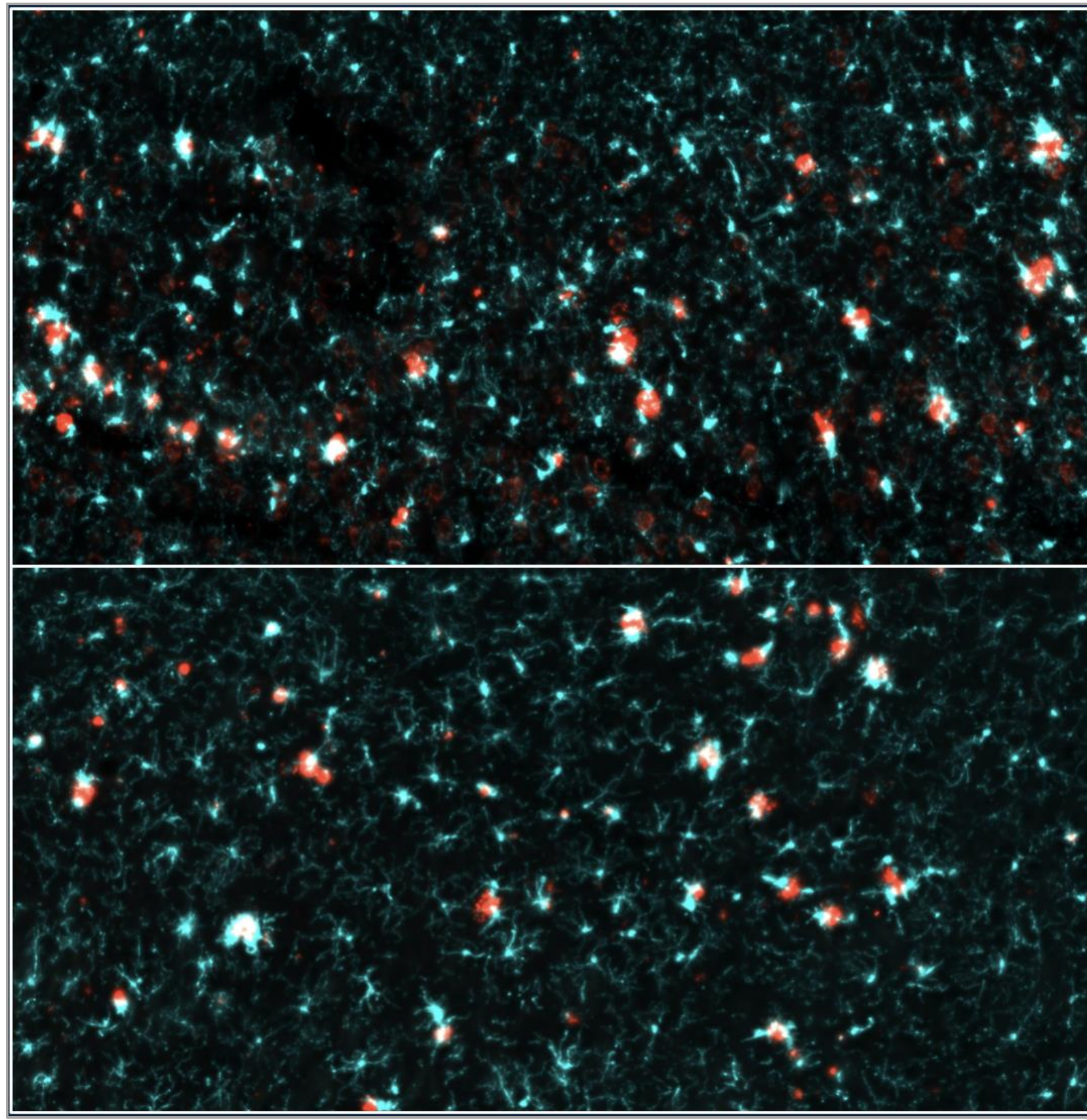


Results & Discussion

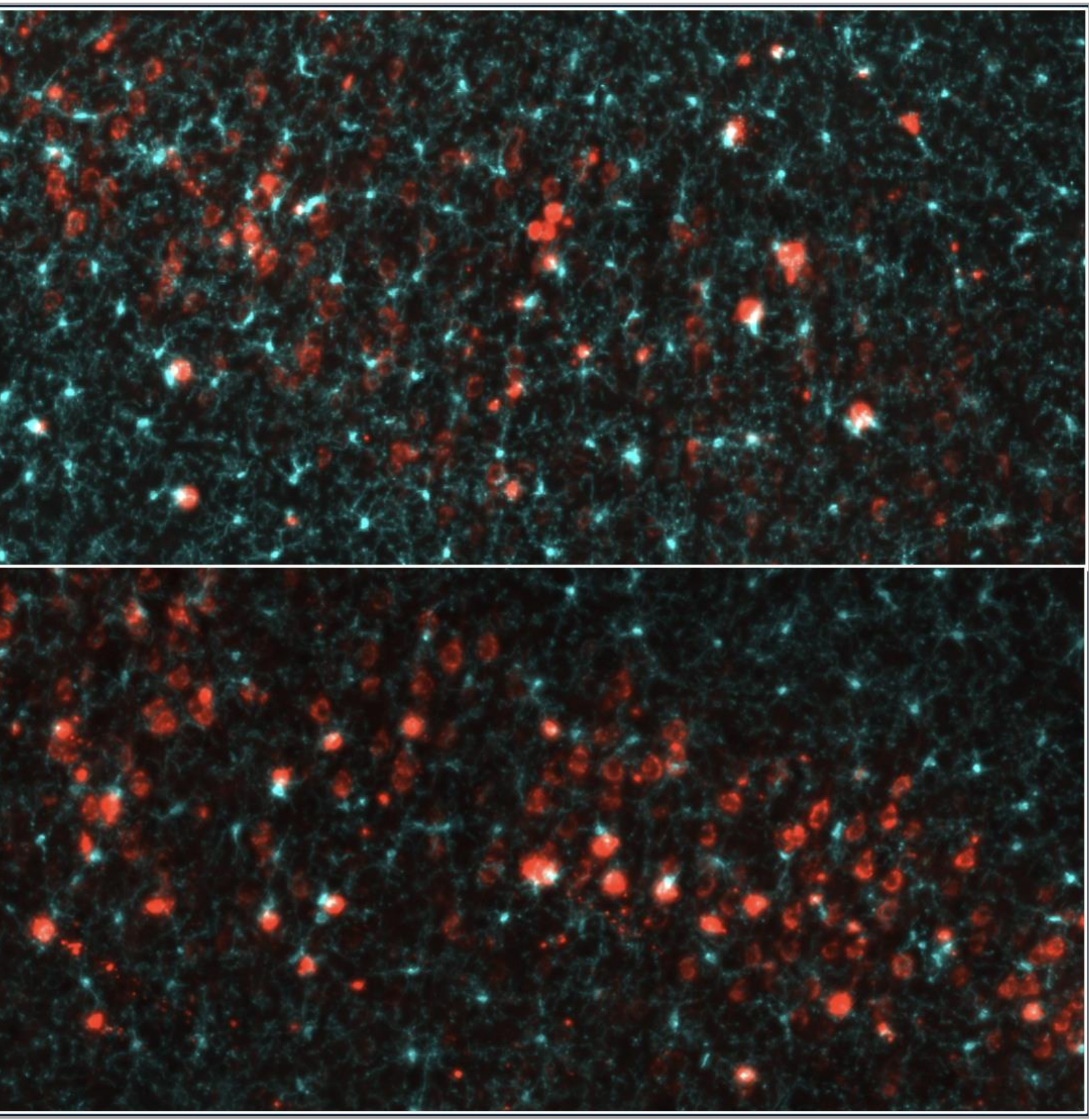
Legend

Red = Aβ-Plaques Blue = Microglia

Healthy-Control



Inflammatory Bowel Disease



5xFAD Cortex - PND 60

- More** plaque-associated **microglia**
- Lower** amyloid-beta **plaque** content

- Less** plaque-associated **microglia**
- Higher** amyloid-beta **plaque** content

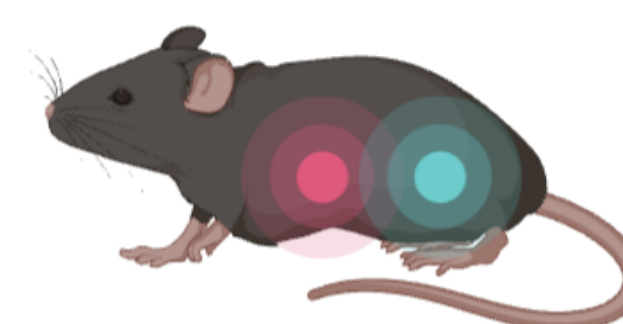
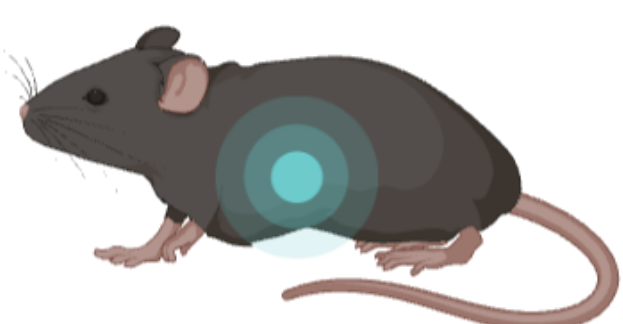
Methodology

Germ-free mice were given either *healthy or IBD human microbiota* from local patients, and their offspring were model systems

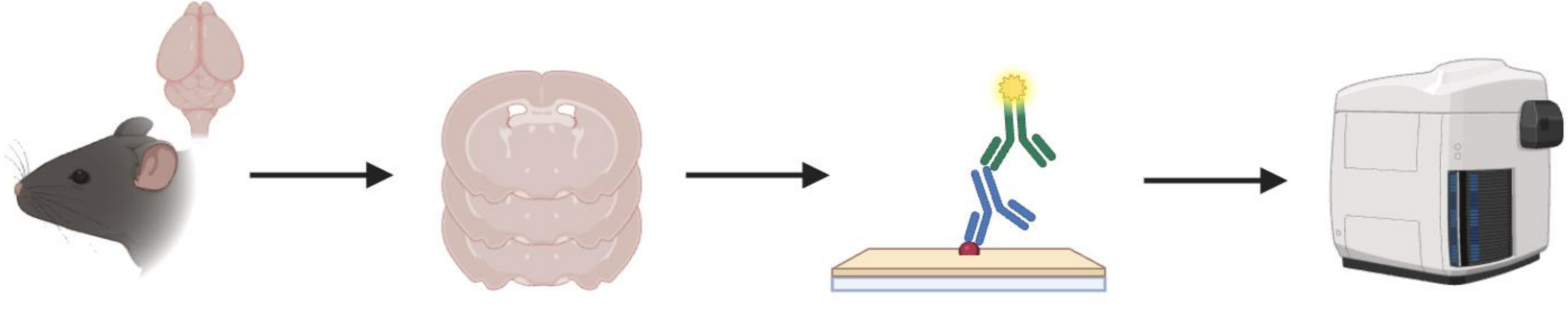
Transgenic 5xFAD mutated mice were representative **AD models**

Model Systems


HC-WT Model IBD-WT Model


HC-AD Model IBD-AD Model

Collect Brains → Section → Immunohistochemistry → Imaging



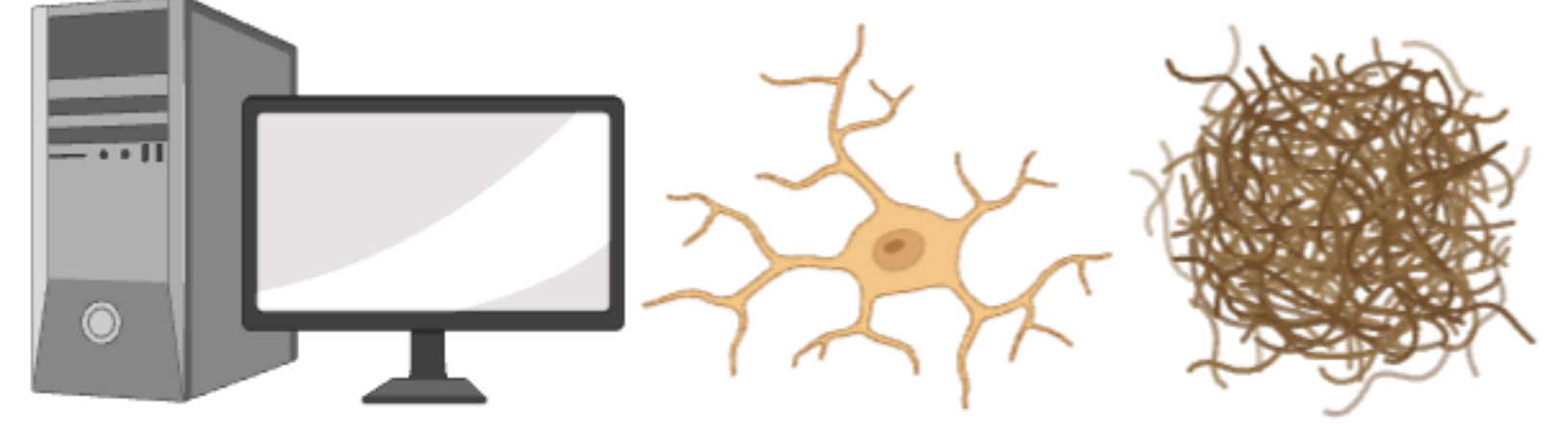
- Mice brains were collected at 60 days old (young adulthood)
- Brains were fixed and cryosectioned coronally
- Microglia** and **amyloid-beta plaque** staining
- Slide scanning and imaging

Graphics created by BioRender

Future Directions & Implications

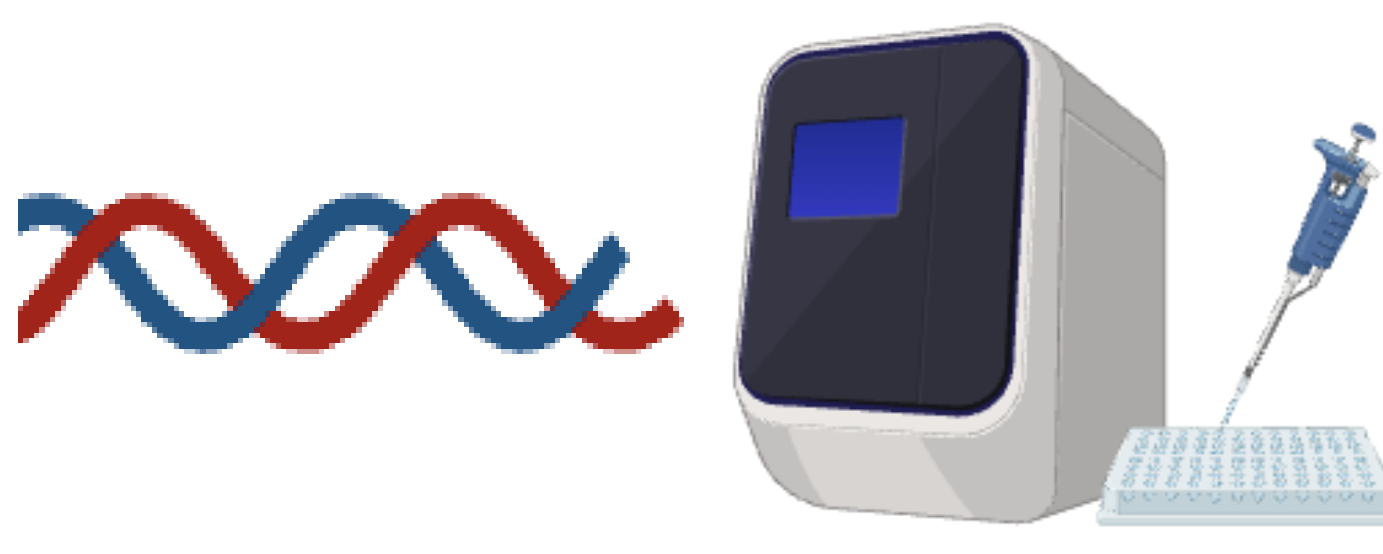
Quantification & Data Analysis

- Microglia**
- Amyloid-beta plaques**
- Statistical tests for significance



Gene Expression Analysis with RT-qPCR

- Inflammatory signalling pathways
- Amyloid & Alzheimer’s genes



Varied Developmental Periods

- 180 days collection
- Onset vs. Progression comparison
- Identify biomarkers for patients



Acknowledgements

Dr. Annie V. Ciernia, Dr. Jordan E. Hamden, and the entire Ciernia Lab have provided invaluable support and mentorship

Research conducted on traditional, ancestral, and unceded territories of the xʷməθkʷəy̓ ʼəm (Musqueam) people

References



Partners

